

TABULATED DELAYED NEUTRON GROUP PARAMETERS FOR MOLTEN SALT REACTORS

BY

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DISSERTATION

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Abstract

Beta-delayed neutron emitters, or delayed neutron precursors (DNPs), are important for safeguards monitoring and safety considerations in reactor transients, such as shutdown decay-heat, reactor controllability, and reactivity changes. In particular, molten salt reactors (MSRs) have DNPs which move from areas of high importance in the core to areas of lower importance in the core and ex-core regions. This causes the effective delayed neutron fraction to decrease, leading to changes in reactivity based on the salt flow rate. Many works have investigated how the effective delayed neutron fraction changes based on the movement of fuel salt in MSRs. What these works do not consider, however, is the effect of fuel salt movement and reprocessing on the DNP group parameters used in the transient simulations. This issue is important to resolve, as the DNP group parameters are used for safety-related calculations as well as in safeguards-related detection simulations.

The DNP group parameters for MSRs can be resolved by first generating concentrations that account for the spatially resolved depletion and reprocessing. Shorter lived DNP group parameters are found via pulse irradiation experiments, which occur at very short time scales relative to molten salt flow rates. The longer lived DNP group parameters vary from traditionally calculated values since they are measured with saturation, or infinite, irradiation experiments which occur at time scales several times longer than the longest DNP half life. These longer time scales mean that thermal-hydraulic, thermochemical, and reprocessing effects are able to affect the DNPs.

The concentrations are combined with delayed neutron emission probability and half life data for each nuclide to construct delayed neutron counts as a function of time, a process described in the literature as the "microscopic" approach. A non-linear least-squares solver is then used in order to generate a χ^2 minimized set of DNP group parameters. These updated MSR DNP group parameters are then used in several safety-related transient and safeguards simulations in the Molten Salt Reactor Experiment in order to demonstrate the difference between the group parameters. This work solves an important problem with the DNP group parameters for use in MSRs by providing updated group parameters and a method for generating them dynamically for a given reactor. Tabulated DNP group parameters for various flow rates and energies are provided in this work for future analyses.

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Chapter 1

Introduction

- Go through history of group parameters (maybe timeline figure?).
- Discuss MSRs.
- "The next step forward for group parameter development is to have group parameters that accurately reflect the physics that occur in fluid-fueled molten salt reactors (MSRs)".
- Discuss methodology at a high level.
- Discuss results at a high level.
- Discuss conclusions of the work at a high level.

Chapter 2

Background and Motivation

- Discuss what motivates this work to set up what needs background information

2.1 Nuclear Safeguards

- Background on nuclear safeguards (proliferation resistance particularly).
- Discuss six proliferation resistance (PR) measures and which are of note for this work.
- special nuclear material (SNM) classification (with table).
- Discuss material attractiveness levels.
- Discuss significant quantities (with bullet points).
- Bring it back to the main point ("This information is important for the current work, as it provides quantities of diverted material to analyze. Generation of group parameters will yield new delayed neutron fractions, which may affect the detection capabilities for material diversion by monitoring delayed neutron fraction")
- Provide example case

2.2 Safety-Related Reactor Transients

- Introduce postulated initiating event (PIE) leading to design basis accidents (DBAs).
- Discuss events International Atomic Energy Agency (IAEA) describes that are of interest.
- Introduce safety related transients
- Describe why they are relevant to this work (higher accuracy in transient simulations, improved safety).

2.2.1 Transient Analysis Methods

- Introduce the different methods for analyzing transients in MSRs

Multi-group diffusion

- Describe multi-group diffusion
- Provide examples of works that use it
- Describe the implementation

Modified point kinetics

- modified point kinetics (MPK): describe, examples, implementations
- delayed point reactor kinetics (DPK): describe, examples, implementations
- IPK: describe, examples, implementations

Quasi-statics

- Describe quasi-statics
- Provide examples of works that use it
- Describe the implementation

2.3 Delayed Neutron Precursors

2.3.1 Overview

- Introduce delayed neutron precursors (DNPs)
- Discuss the impact on reactor period and effect on transients
- Discuss challenge in spatial treatment
- Discuss why DNPs are important to model (over-conservatism increases cost, difficulty in licensing; while under-conservatism increases risk)

2.3.2 Measurement

- Overview of macroscopic and microscopic methods

Macroscopic Measurement of Delayed Neutron Precursors

- Describe the macroscopic approach in detail, giving examples from the literature
- Describe infinite and pulse irradiation
- State how the generated data is used (point to Section 2.3.3)
- Show image of experimental setup example
- Discuss setup and challenges

Microscopic Measurement of Delayed Neutron Precursors

- Describe the microscopic approach in detail with examples from the literature
- Describe experiments used to collect the data
- Show an image of experimental setup example
- Describe why microscopic approach is useful
- Talk about IAEA database and how it can be used

2.3.3 Group and Kinetics Parameters

Overview

- Re-introduce DNP groups and their purpose (reference Section 2.3.2 for experimental discussion)
- Discuss DNP group parameters (yield, half-life, delayed neutron fraction, spectra, total yield)
- Discuss challenges/issues with groups (mostly historical)
- Discuss usefulness of DNP in kinetic parameter calculations
- Introduce k_{eff} and $k_{infinity}$
- Introduce β
- Introduce β_{eff}
- Introduce Λ and Λ_{eff}
- Discuss adjoint flux importance

Calculation of group parameters

- Introduce exact calculation of delayed neutron counts
- Discuss fit delayed neutron counts measured by an irradiation (and implicit simplifications)
- Discuss pulse vs saturation irradiation (as well as combination)
- Show how saturation is used to get average number of neutrons per fission
- Discuss over-determined non-linear least squares methodology with equations
- Discuss differences if there are multiple fissioning nuclides with equation
- Discuss energy spectra

Calculation of group concentrations and spectra

- Relate concentration calculation to energy spectra calculation
- Discuss calculating group concentrations from individual nuclide data
- Discuss different methods (half-lives and fractional contribution)
- Show table of half-lives and fractional contribution data
- Discuss each approach in more detail with equations

Calculation of kinetics parameters

- Discuss beta calculation
- Discuss ν_d calculation
- Show how beta relates to β
- Discuss Λ (prompt neutron lifetime and k) (calculation with PRK and time-dependent Monte Carlo)
- Discuss beta eff
- Discuss Λ_{eff}
- Discuss methods to calculate betaeff and Λ_{eff} in the literature

2.4 Molten Salt Reactor Depletion

2.4.1 Overview

- Introduce MSRs with examples from literature.
- Discuss treatment of reprocessing in depletion simulations of MSRs (table from masters)
- These models do not include spatial resolution (beyond flux scaling)
- Discuss why spatial resolution is important for this work

2.4.2 Depletion Simulations

- Introduce depletion and the Bateman equation
- Discuss the matrix form
- Discuss numerical methods for solving the matrix form

Stationary fuel

- Write Bateman equation in a form comparable for the circulating fuel form
- Discuss approaches used when solving depletion for stationary fuel

Circulating fuel

- Introduce Bateman with advection and diffusion
- Describe how circulating fuel depletion is challenging
- Discuss different approaches to reduce computational cost

Chapter 3

Methodology

Maybe I could also add something in the preliminary work on tortuosity/diffusion/turbulent mixing (depending on if it could be implemented into the simplified model). Another consideration could be using a sinusoidal flux shape for the irradiation history (small at inlet, becomes larger, then decreases as it moves towards the outlet) (could be interesting to see if this has a non-negligible effect on the results).

3.1 Tools

- Provide a broad overview of tools needed

3.1.1 Python Scripts

- Discuss various packages used in Python scripts
- Describe broadly what the Python scripts do

3.1.2 OpenMC

- Describe OpenMC
- Describe specifically how this work uses OpenMC

3.1.3 MOOSE Apps

- Describe MOOSE

Moltres

- Describe Moltres
- Describe how it is used in this work

3.2 Tabulation

- Discuss "reactor agnostic" approach (ex-core time, in-core time, and mixing rate (maybe diffusion/tortuosity?))
- Further describe the approach in more detail
- Discuss chemical removal tabulation (and non-inclusion of refueling effects)
- Discuss Molten Salt Breeder Reactor (MSBR) reprocessing scheme (include table from [3])
- Discuss in general the various tabulation parameters and simplifications made to reduce the total number of parameters

3.2.1 Simplified Model

- Introduce simplified model (used to understand general trends that can be expected from advection) (Appendix for derivation of non-linear least-squares formatted equation)
- Show flow diagram and discuss (several paragraphs to fully explain)
- Discuss specific considerations for OpenMC simulation (self-shielding)
- Discuss specifics of OpenMC model
- Discuss consistent source particle rate between simulations (with equation)

3.3 Resolved Delayed Neutron Counts

- Introduce coming subsections which will offer more details on what the OpenMC model does (equations)

3.3.1 Delayed Neutron Precursor Concentrations

- Discuss density variations in OpenMC model
- Discuss energy spectra in OpenMC model (mention photofission)
- Describe source distribution in the fixed-source problem and reasoning
- Describe how concentrations are collected and passed along

3.3.2 Concentrations to Counts

- Describe formation of concentrations as a function of time using typical approach in the literature
- Discuss using OpenMC decay with small time steps to track decay parents of precursors (comparison in Section 4.4.3)
- Show recreated equation of delayed neutron count rate over time
- Discuss how the number of DNPs is where OpenMC concentrations exist and IAEA database data exists (should underestimate delayed neutron yield if there is missing data)

3.4 Delayed Neutron Precursor Group Reconstruction

- Conversion of counts into group parameters
- Discuss the time values used for generating results (see 4.3.1)
- Covariance of non-linear least-squares solve
- Linearization, normalization, and tolerances
- Sample self-multiplication
- Uncertainty tracking and minimization of χ^2
- Combination of saturation and pulse results from Brady [1] and what this work does

Chapter 4

Preliminary Results

- Describe the specific modeling choices made to reduce tabulations.
- Describe modeling choices for non-tabulated parameters (temperature, time nodes, decay daughter inclusion, density)

4.1 Simplified Model Refinement

4.1.1 Temperature

4.1.2 Target Density

4.1.3 Total Irradiation Time

4.1.4 Total Decay Time and Decay Time Steps

The bulk of the computational cost comes from writing statepoint files; sufficient resolution is required to ensure a smooth decay curve.

4.2 Tabulation Refinement

Discuss effect of varying reprocessing; determine what differences arise from inclusion of larger cycle time elements (if any). (might need to adjust methodology section based on these results). Scale the reprocessing terms differently to see their effects (see if each can be scaled the same way).

4.2.1 Large Cycle Time Analysis

4.2.2 Reprocessing Rate Scaling

4.2.3 Residence Time Variation

4.3 Parameter Fit Refinement

4.3.1 Time Evaluations

4.3.2 Delayed Neutron Count Uncertainties

4.3.3 Group Parameter Uncertainties (?)

4.4 Group Parameter Comparisons

4.4.1 Comparison with Documented Results

4.4.2 Comparison of Current Results

4.4.3 Comparison with Decay Chain Daughters

Use $N(t)$ from SaltProc with linear interpolation.

4.5 Spatially Resolved Depletion Effects on Precursors

Net delayed neutron yield; group abundances; group decays; comparison of most impactful nuclides (?)

Include polynomial fits (figures) showing how group parameters vary based on tabulated parameters (in-core, ex-core, chemical removal scaling, etc.)

4.6 Verification

4.6.1 Simplified Model

4.6.2 No Molten Properties Comparisons

No flow, reprocessing, thermochemistry, spatial depletion should equal expected results.

4.6.3 Literature Comparison

Should match other Griffin/Pronghorn results. Maybe compare with Moltres/Quasimolto MSRE pump start-up/coast-down. Benchmark Sun Myung talked about?

4.7 Validation

4.7.1 MSRE Rod Insertion

Sun Myung integral rod worth experimental results were slightly larger than Moltres results, maybe using updated group parameters would bring it closer?

Chapter 5

Proposed Work

5.1 Higher-fidelity Model

5.1.1 Group Generation

- Moltres can be used to perform depletion (Nathan Glaser's work)
- Can have spatially resolved irradiation using this functionality (while accounting for thermal-hydraulics)
- Flow analyses can be conducted (channel flow, open core, different fuel diffusion coefficients)
- Verify if simplified model can be used to generate group parameters (if so, when; else why not) (reprocessing from ex-core vs OpenMC dimensionless reprocessing)
- Compare computational cost to accuracy

5.1.2 Group Evaluation

- Moltres takes in group parameters to evaluate reactor transients
- Various safety and safeguards scenarios can be investigated
- Scenarios where the difference between static and flowing group fits leads to incorrect modeling can be discussed

5.2 DNP Spectra

- [2] discusses why it's important in section 6
- Discuss how spectra could be resolved
- Can also be compared between higher fidelity model

5.3 Resolve DNP spectra

5.4 Uncertainty Quantification

Compare with Kozlowski paper Using sensitivity to flow and reprocessing. Uncertainty in half lives and emission probabilities is small. Neglect uncertainty in concentrations from OpenMC (cross sections, fission yields, half lives, transmutations, decay yields, etc.)?

5.5 Other extensions

Diffusion/mixing effect approximated in simplified model? Photoneutrons - investigate Spatially resolved flux - consider, Specific neutron spectra (mentioned in methodology) Thermochemical effects - different elements will have different chemical removal rates, which will affect results (Sam knows more about this and could help me figure out if there would be something here worth investigating)

Appendix A

Appendix A: Derivation of Simplified Flowing Irradiation Delayed Neutron Counts for Equal Residence Times

This derivation begins with the Equations (AA.1) and (AA.2) provided by Wilson and England [4]:

$$R(t) = \sum_{i=1}^n a_i \lambda_i e^{-\lambda_i t} \quad (\text{AA.1})$$

$$n_d(T, t) = \bar{\nu}_d \int_0^T F(\tau) R(T - \tau + t) d\tau \quad (\text{AA.2})$$

The terms in Equation (AA.1) are discussed in Section 2.3.3, and the terms not discussed in the text in Equation (AA.2) are defined as follows:

- T is the length of the fission history
- t is the decay time from the end of the fission history
- $F(\tau)$ is the fission history (in fissions per second)
- $\bar{\nu}_d$ is the average number of delayed neutrons per fission event

Because the results are normalized to a single delayed neutron, the $\bar{\nu}_d$ term is divided out.

To derive an analytical solution to this problem for the simplified approach where the in-core and ex-core residence times, τ_{in} and τ_{ex} , are equivalent, an approximation of the fission history must be made. For this derivation, τ_{res} will be used to represent both the in-core and ex-core residence times. First, the fission history is split into a shape and amplitude function, shown in Equation (AA.3). This essentially just simplifies any further equations such that only the shape needs to be considered.

$$F(t) = F_0 \psi(t) \quad (\text{AA.3})$$

Because the simplified model's fission history is a positive square wave, analytical integration is not feasible. Instead, a Fourier decomposition of the square wave into a sum of sine waves makes the problem far more approachable. For numerical integration this is not an issue, but an analytical solution is useful for setting up the least squares matrices.

$$\psi(t) = \frac{1}{2} + \frac{2}{\pi} \left(\sin\left(\frac{\pi t}{\tau_{res}}\right) + \frac{1}{3} \sin\left(\frac{3\pi t}{\tau_{res}}\right) + \dots \right) \quad (\text{AA.4})$$

Plugging this in and separating the integral gives Equation (AA.5) for group i and for k Fourier terms.

$$\begin{aligned} n_{d,i}(T, t) = & \frac{F_0 a_i \lambda_i}{2} \int_0^T e^{-\lambda_i(T-\tau+t)} d\tau \\ & + \frac{2F_0 a_i \lambda_i}{\pi} \int_0^T \sin\left(\frac{\pi \tau}{\tau_{res}}\right) e^{-\lambda_i(T-\tau+t)} d\tau \\ & + \dots \\ & + \frac{2F_0 a_i \lambda_i}{(2k+1)\pi} \int_0^T \sin\left(\frac{(2k+1)\pi \tau}{\tau_{res}}\right) e^{-\lambda_i(T-\tau+t)} d\tau \end{aligned} \quad (\text{AA.5})$$

Solving the integrals and simplifying yields Equation (AA.6), where each integrated Fourier term generates four terms added to the sum.

$$\begin{aligned} n_{d,i}(T, t) = & \frac{F_0 a_i e^{-\lambda_i t} (1 - e^{-\lambda_i T})}{2} \\ & + \dots \\ & + \frac{2F_0 a_i \lambda_i^2 \tau_{res}^2 e^{-\lambda_i t}}{((2k+1)\pi) ((2\pi k + \pi)^2 + \lambda_i^2 \tau_{res}^2)} \sin\left(\frac{\pi(2k+1)T}{\tau_{res}}\right) \\ & + \frac{2F_0 a_i \lambda_i \tau_{res} e^{-\lambda_i t} e^{-\lambda_i T}}{((2\pi k + \pi)^2 + \lambda_i^2 \tau_{res}^2)} \cos\left(\frac{\pi(2k+1)T}{\tau_{res}}\right) \\ & + \frac{2F_0 a_i \lambda_i \tau_{res} e^{-\lambda_i t} e^{-\lambda_i T}}{((2k+1)\pi) ((2\pi k + \pi)^2 + \lambda_i^2 \tau_{res}^2)} 2\pi k \\ & + \frac{2F_0 a_i \lambda_i \tau_{res} e^{-\lambda_i t} e^{-\lambda_i T}}{((2k+1)\pi) ((2\pi k + \pi)^2 + \lambda_i^2 \tau_{res}^2)} \pi \end{aligned} \quad (\text{AA.6})$$

Similar to [4], the irradiation time is taken to the limit as it approaches infinity, though only for the non-periodic terms. By irradiating for several half-lives of time for each DNP, where the longest half-life is roughly one minute, then the irradiation time can be taken as infinite. There are two reasons this limit is taken. Firstly, a pulse irradiation with circulation will not provide any unique results when compared to traditional data which already exists. Additionally, this allows the equation to be simplified, as shown in Equation (AA.7). Only the sine term does not go to zero as the irradiation time goes to infinity, so it is the only term significant for saturation irradiation in the simplified model.

$$\begin{aligned}
n_{d,i}(T, t) &= \frac{F_0 a_i e^{-\lambda_i t}}{2} \\
&+ \dots \\
&+ \frac{2F_0 a_i \lambda_i^2 \tau_{res}^2 e^{-\lambda_i t}}{((2k+1)\pi) \left((2\pi k + \pi)^2 + \lambda_i^2 \tau_{res}^2 \right)} \sin \left(\frac{\pi(2k+1)T}{\tau_{res}} \right)
\end{aligned} \tag{AA.7}$$

The final assumption to make is that the irradiation time, T , is an integer multiple of the residence times, τ_{res} . This assumption makes the sine term 0 for every value of k , which means that the result simplifies to (AA.8).

$$n_d(t) = \frac{F_0}{2} \sum_{i=1}^n a_i e^{-\lambda_i t} \tag{AA.8}$$

This is very similar to the saturation irradiation delayed neutron count equation, shown in Equation (AA.9).

$$n_d(t) = F_0 \sum_{i=1}^n a_i e^{-\lambda_i t} \tag{AA.9}$$

Because of the initial assumption made, which was that half the time is in-core and half ex-core, the average fission rate is half of the fission rate scaling term, F_0 . For both of the cases of constant irradiation and half in-core and half ex-core, the fission term in front is simply the average fission rate over the irradiation time.

References

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